

Semiconductor Design Infrastructure Handbook

Deployment, Configuration, and Administration
of Linux-based EDA workstations.

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Abstract

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This manual covers the procedures required to deploy, provision, operate, maintain, and recover a RHEL-based semiconductor CAD workstation environment. It is intended to serve as an operational reference for administrators responsible for preparing workstations, installing EDA tools, configuring user access, and maintaining a reliable design environment over time.

Specifically, this manual covers the following areas:

- Operating system deployment for RHEL workstations, including installation planning and baseline system configuration.
- Workstation provisioning, including package installation, network readiness, storage preparation, and post-installation setup.
- EDA installation workflows for supported semiconductor design tools and their required dependencies.
- Licensing configuration, including license server connectivity, environment variables, and validation steps.
- Shell environments for CAD users, including startup files, module or tool setup scripts, and reproducible environment initialization.
- User management tasks, including account creation, group membership, access control, and basic permission practices.
- Troubleshooting procedures for common operating system, EDA tool, license, shell, and user-access issues.
- Recovery procedures for failed installations, broken environments, workstation rebuilds, and service restoration.
- Maintenance workflows, including updates, backups, configuration audits, cleanup tasks, and periodic validation of the CAD environment.

This manual does not cover semiconductor design methodology, RTL design, verification strategy, physical design theory, circuit design, device modeling, or EDA tool usage from a design-engineering perspective. It also does not define institutional IT policy, network architecture outside the workstation environment, procurement procedures, vendor contract negotiation, or license entitlement management beyond the technical configuration needed for workstation operation.

Preface

Modern civilization is built upon silicon. From mobile communication systems and medical instrumentation to aerospace control systems, artificial intelligence accelerators, and high-performance computing platforms, nearly every contemporary technological system is ultimately dependent upon integrated circuits containing billions of transistors operating at nanometer-scale geometries. The design and verification of such devices would be computationally impossible without the development of sophisticated Electronic Design Automation (EDA) frameworks capable of transforming abstract circuit descriptions into physically manufacturable semiconductor layouts. Over the past several decades, EDA has evolved from a collection of isolated drafting and simulation utilities into a deeply interconnected computational ecosystem comprising schematic capture systems, hardware description language toolchains, simulation engines, parasitic extraction frameworks, physical verification suites, and advanced layout environments operating across distributed compute infrastructure. As semiconductor complexity increased, the underlying computing environments required to support these workflows evolved alongside them, giving rise to highly specialized workstation architectures optimized for reliability, reproducibility, performance, and long-term maintainability.

Contemporary semiconductor design environments are overwhelmingly deployed on Linux-based operating systems due to their stability, scalability, scripting flexibility, network integration capabilities, and compatibility with industry-standard EDA toolchains. Unlike general-purpose desktop operating systems, enterprise Linux distributions provide deterministic package management, robust permission models, multi-user infrastructure support, and predictable system behavior required for large-scale engineering workflows. Among these distributions, Red Hat Enterprise Linux (RHEL) and its derivatives have emerged as the de facto industry standard within both commercial semiconductor organizations and academic VLSI laboratories owing to their long-term support lifecycle, binary compatibility guarantees, enterprise-grade tooling ecosystem, and extensive vendor certification support from major EDA providers. Consequently, the deployment methodology described throughout this manual is centered around a standardized RHEL-based semiconductor CAD environment designed to support scalable, reproducible, and maintainable VLSI design workflows across multiple engineering workstations.

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Part I

SYSTEM SETUP

Operating System Deployment and RHEL Workstation Initialization

01

RHEL Installation and Setup

1.1 What is EDA and why is it used?

Electronic Design Automation (EDA) refers to the collection of software tools, computational frameworks, and engineering workflows used in the design, simulation, verification, and physical implementation of integrated circuits and semiconductor systems. Modern semiconductor devices contain millions to billions of transistors operating at extremely small geometries, making manual design methodologies computationally impractical and operationally impossible. EDA tools therefore serve as the foundational infrastructure enabling engineers to transform conceptual circuit designs into manufacturable silicon devices.

Contemporary EDA environments encompass a wide range of specialized applications including schematic capture systems, Hardware Description Language (HDL) development environments, circuit simulators, synthesis tools, physical layout editors, parasitic extraction frameworks, timing analysis engines, and design-rule verification systems. These tools collectively support the complete semiconductor development lifecycle, from behavioral modeling and functional verification to physical layout generation and fabrication validation.

The increasing complexity of Very Large Scale Integration (VLSI) systems has made EDA indispensable within both academic and industrial semiconductor design workflows. Beyond improving productivity, EDA enables precision, scalability, repeatability, and verification accuracy that would otherwise be unattainable through conventional design methodologies. As a result, modern semiconductor engineering is fundamentally dependent upon robust EDA infrastructure and the computational environments that support it.

1.2 Why Enterprise Linux Anchors Modern EDA?

Modern EDA toolchains impose demanding requirements on the underlying operating system environment, including stability, deterministic behavior, scripting flexibility, network interoperability, multi-user access control, and long-term software compatibility. Consequently, semiconductor design organizations and research laboratories overwhelmingly deploy EDA infrastructure on Linux-based operating systems rather than general-purpose desktop platforms.

Linux provides a highly modular and administratively transparent environment capable of supporting complex engineering workflows involving distributed compute resources, centralized licensing systems, shared storage infrastructure, remote administration, and automated deployment methodologies. Native shell scripting capabilities, package management systems, process control utilities, and network tooling further enable scalable workstation administration and reproducible environment configuration across large laboratory deployments.

Among available Linux distributions, enterprise-grade platforms such as **Red Hat Enterprise Linux (RHEL)** and its derivatives have become the de facto standard within semiconductor design environments. Their widespread adoption is driven by long-term support lifecycles, predictable release stability, extensive vendor certification, and broad compatibility with commercial EDA applications. Most major semiconductor

tool vendors officially validate and support their software primarily on enterprise Linux environments, making RHEL-based systems the preferred foundation for reliable and maintainable CAD infrastructure deployment.

Accordingly, the workstation architecture described throughout this manual is built upon a standardized RHEL-based environment designed to provide operational consistency, scalability, and long-term maintainability for semiconductor design laboratories and VLSI research workflows.